



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	INTRODUCTORY GENERAL RELATIVITY				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Special relativity. 2. Tensors.				
Course Summary	This course introduces Einstein's general theory of relativity in a quantitative manner. The mathematical foundations required are developed before discussion of the theory. The mathematical concept behind black holes is also introduced.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Review of tensors, as well as understand tensor calculus	R, U	C	Instructor-created exams / Quiz
CO2	Understand the metric tensor and how curved spacetime is described by the metric tensor.	U, Ap	C	Instructor-created exams / Quiz
CO3	Understand Christoffel's symbols and the Riemann–Christoffel curvature tensor.	U, Ap	C	Instructor-created exams / Quiz / Home Assignments.
CO4	Understand how Parallel displacement can be used to detect curvature.	Ap	C	Instructor-created exams / Quiz
CO5	Understand equivalence principle and principle of general covariance to arrive at Einstein's equations	U	C, F	Instructor-created exams / Quiz
CO6	Understand the basic mathematical theory behind black holes.	U	C, F	Instructor-created exams / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	INTRODUCTION		10	15
	1	Introduction, Relativity as a co-ordinate symmetry, GR as a gravitational field theory.	1	

	2	Line element, Riemannian space, transformation of coordinates.	2	
	3	Contravariant and covariant vectors, summation convention.	1	
	4	The metric, the metric as a tensor, contravariant, covariant, and mixed tensors.	3	
	5	Multiplication of tensors—inner and outer products, contraction, fundamental tensors: $g_{\mu\nu}$, $g^{\mu\nu}$, and g_{μ}^{ν} , raising and lowering of indices.	3	
1. Section 1.1 (1.1.4 not required), 1.2 (upto and including 1.2.3. In 1.2.3, only Tensor formalism is required) from Book 1 2. Section 2.1-2.11 of Book 2.				
II	COVARIANT DIFFERENTIATION AND THE METRIC TENSOR		10	15
	6	Manifolds, the partial derivative of a tensor.	3	
	7	Covariant differentiation and the affine connection.	3	
	8	Christoffel's 3-index symbols and their transformation law.	2	
	9	Geodesics, covariant differentiation of vectors, covariant derivatives of tensors.	2	
Section 2.12-2.15 of Book 2				
III	CURVATURE TENSOR AND EINSTEIN'S FIELD EQUATIONS		21	25
	10	Riemannian coordinates, Riemann–Christoffel curvature tensor.	3	
	11	Symmetries and anti-symmetries of curvature tensor.	3	
	12	Number of independent components of the curvature tensor $R_{\lambda\mu\nu\sigma}$	1	
	13	The Bianchi Identities, The Ricci tensor	3	
	14	The Einstein tensor, the condition for flat space-time.	4	
	15	The equivalence principle.	3	
	16	The principle of general covariance.	1	
	17	Heuristic derivation of Einstein field equations.	2	

	18	Fundamental hypotheses and postulates of general relativity.	1	
Sections 2.18-2.21, 2.23-2.24, 3.1, 3.2 (3.2.1-3.2.3 not required), 3.5, 3.6 (final equation only) of Book 2				
IV	SCHWARZSCHILD SOLUTION AND BLACK HOLES		7	15
	19	Introduction.	1	
	20	A static, spherically symmetric space–time (general idea only, derivation not required),	2	
	21	The Schwarzschild line-element (general idea only, derivation not required), Schwarzschild Singularity.	2	
	22	Schwarzschild Black Holes—Singularities.	2	
Section 4.1-4.3 (upto and including 4.3.1), 7.1,7.2 of Book 2				
V	OPEN ENDED MODULE: BASIC COSMOLOGY		12	
		The cosmological principle, homogeneity and isotropy, different types of curvature, the Robertson-walker metric, Friedmann equation, three crucial tests of the general theory of relativity.		
	References 4-5			
Books and References:				
1. Relativity Gravitation & Cosmology 2 nd edition, Ta-Pei Cheng, Oxford University Press, 2010 (Book 1)				
2. General Theory of Relativity, S. P. Puri, Pearson India, 2013 (Book 2)				
3. An Introduction to Relativity by Jayant V. Narlikar, Cambridge University Press				
4. Introduction to Cosmology by Jayant V. Narlikar, Cambridge University Press				
5. Gravity: An Introduction to Einstein's General Relativity Hardcover by James Hartle				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PS O4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	1	2	0	2	2	3	2	3	1	2	3	3
CO 2	3	1	2	0	2	2	3	2	3	1	2	3	3
CO 3	3	1	2	0	2	2	3	2	3	1	2	3	3
CO 4	3	1	2	0	2	2	3	2	3	1	2	3	3
CO 5	3	1	2	0	2	2	3	2	3	1	2	3	3
CO 6	3	1	2	0	2	2	3	2	3	1	2	3	3

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	